

The QSC Atlas evaluative layer: methodology, a run across all countries, and the build brief

Six criteria, scored from the recorded data, with a detailed UI specification for Claude Code Working note, June 2026. Companion to Extending the QSC Atlas and the QSA methods synthesis. [Author and affiliation]

Summary. This note operationalises the evaluative layer, runs it across every country in the dataset, and specifies how to build it. Part A defines the six criteria as eleven sub-criteria, each scored 0 to 2 against stated descriptions, with the verification and reconciliation rules that govern scoring. Part B gives a first-pass run for every coloured country in the dataset, derived mechanically from the recorded Atlas fields by the rules in the note, with confidence carried and reviewer reconciliation pending. Part C is the detailed build brief for the credibility instrument, written for Claude Code. No score is shown unless it traces to a recorded field or source; an unsupported sub-criterion reads as not assessed.

Part A. Methodology

1. What the layer reposes on

The layer adapts the comparative policy-evaluation method that Paglieri et al. (2025) built for national quantum strategies from OECD and World Economic Forum practice, the one method in the assessment corpus designed at the national scale. Vance (2025) sharpens it by separating evaluation into process, impact and cost-benefit, so the layer judges effectiveness and impact by design rather than by outcomes an emerging field cannot yet show. Two of the sub-criteria draw their content from the CEPS governance recommendations, the inventory duty and the capability checks (Pupillo et al., 2025), and the threat articulation criterion from the CEPS governance report (Pupillo et al., 2023). The whole sits inside governance of expectations, which treats this transition as provisional because the threat's timing is unknowable (Budde and Konrad, 2019), so the layer scores credibility under uncertainty and expects the score to move.

2. The scoring scheme

Each of the six criteria is scored through one to three sub-criteria, eleven in total. Every sub-criterion takes one of three values: 0 absent, 1 partial, 2 present, against the description in Table 1. A criterion's score is the mean of the sub-criteria that can be assessed, giving a value from 0 to 2 on each of six axes. The layer reports the six axes as a profile, not a single league-table number, because a composite would hide the disagreement between axes that is often the finding. Three bands label each axis: nascent (0 to 0.7), developing (0.8 to 1.4), established (1.5 to 2).

Criterion	Sub-criterion	Score: 0 / 1 / 2	Reposes on
Relevance	R1 Transition plan	none / announced or partial / published roadmap with milestones	Paglieri et al. (2025); NIST (2022)
	R2 Threat articulation	absent / generic / names the harvest-now-decrypt-later exposure	Paglieri et al. (2025); Pupillo et al. (2023)
Coherence	C1 Policy integration	isolated / referenced in wider cyber and S&T	Paglieri et al. (2025), after OECD

	C2 Consultation	policy / embedded in it none / narrow / broad consultation across government, industry and academia	Paglieri et al. (2025), after WEF
Effectiveness	E1 Milestones	none / indicative / dated and sector-specific	Vance (2025); Paglieri et al. (2025)
(by design)	E2 Inventory duty	none / recommended / mandated (inventory or CBOM)	Pupillo et al. (2025), GR3; Vance (2025)
	E3 Measurement	none / ad hoc / a monitoring framework or indicators	Paglieri et al. (2025), after OECD
Efficiency	Ef1 Proportionate mix	mismatched / partial / instrument strength matched to objective	Paglieri et al. (2025), after OECD
Governance quality	G1 Mandate and authority	none / a named lead / a mandated lead with a steering function	Paglieri et al. (2025); Pupillo et al. (2025), GR9
	G2 Capacity and skills	none / acknowledged / a skills or capability programme	Pupillo et al. (2025), GR10
Impact (structured-for)	I1 Exposure-reducing adoption	none / baseline referenced / baseline adopted with hybrid and agility	Vance (2025)

Table 1. The eleven sub-criteria, their scoring descriptions, and the model each reposes on.

3. Verification, reconciliation and confidence

Three rules keep the scores defensible. No sub-score is recorded unless it traces to a sourced instrument already held on the country's Atlas profile, or a source added for the purpose; a sub-criterion with no supporting source is left unscored and reads as not assessed, never a default value, and its axis is the mean of the sub-criteria that could be assessed. Second, scoring is reconciled: a second senior reviewer scores each country independently and the two reconcile before a profile is shown to a client, the validation step Paglieri et al. (2025) build into their method. Third, the Atlas's confidence grade is carried onto each axis, so a country assessed on thin evidence reads as tentative. Because the frame is governance of expectations, a score is a reading at a date, not a verdict, and the profile carries the date it was last verified.

Part B. The run across all countries

The table below is a first-pass run for every coloured country, produced by applying the rules in Table 1 mechanically to the fields each profile already records. It is a starting point for review, not a finished assessment: the scores inherit the confidence grade of the underlying profile, the reviewer reconciliation in section 3 is still pending, and the data-thin sub-criteria are scored only where a field supports them. Section 4 records the derivation so each score is traceable.

4. How each score is derived from the recorded fields

R1 and E1 read the migration timeline and target year. R2 reads whether a binding instrument exists, with a softer instrument or a recorded camp counting as partial. C1 reads whether EU instruments or two or more national instruments are present. C2 reads the breadth of recorded governmental and standards bodies. E2 reads the regulation and obligation text for an inventory or CBOM duty, with a binding regime counting as partial. E3 reads the text for a monitoring or audit duty and is left

unscored where none appears. Ef1 reads the legal status. G1 reads the lead authority and steering bodies. G2 reads the text for a skills or capability programme and is left unscored where none appears. I1 reads the standards role where present, and otherwise the dominant-process camp together with the hybrid stance and the recorded standards. A dash means the axis could not be assessed from the recorded fields.

Country	Rel	Coh	Eff	Effi	Gov	Imp	Conf.
Australia	2.0	2.0	1.5	2.0	2.0	2.0	Medium
Austria	2.0	1.5	1.5	2.0	0.0	2.0	High
Bahrain	0.5	0.5	0.0	0.0	1.0	1.0	Medium
Belgium	2.0	2.0	1.5	2.0	2.0	2.0	Medium
Brazil	2.0	2.0	1.5	2.0	2.0	2.0	Medium
Brunei	0.5	0.5	0.0	0.0	1.0	1.0	Low
Bulgaria	2.0	1.5	1.5	2.0	1.0	2.0	High
Canada	2.0	2.0	1.5	2.0	2.0	2.0	Medium
Chile	0.5	0.5	0.0	0.0	2.0	1.0	Medium
China	1.0	1.5	0.5	1.0	1.0	2.0	Medium
Croatia	2.0	1.5	1.5	2.0	1.0	2.0	High
Cyprus	2.0	1.5	1.5	2.0	1.0	2.0	High
Czechia	2.0	2.0	1.5	2.0	2.0	2.0	Medium
Denmark	2.0	2.0	1.5	2.0	2.0	2.0	High
Djibouti	0.5	0.5	0.0	0.0	1.0	1.0	Low
Estonia	2.0	1.5	2.0	2.0	0.0	2.0	High
Finland	2.0	2.0	2.0	2.0	2.0	2.0	Medium
France	2.0	2.0	1.5	2.0	2.0	2.0	Medium
Germany	2.0	2.0	1.5	2.0	2.0	2.0	Medium
Greece	2.0	1.5	1.5	2.0	2.0	2.0	Medium
Hungary	2.0	2.0	1.5	2.0	2.0	2.0	Medium
India	2.0	2.0	2.0	2.0	2.0	2.0	Medium
Indonesia	0.5	0.5	0.0	0.0	2.0	1.0	Medium
Ireland	2.0	1.5	1.5	2.0	1.0	2.0	High
Israel	2.0	2.0	2.0	2.0	2.0	2.0	Medium
Italy	2.0	2.0	1.5	2.0	2.0	2.0	Medium
Japan	2.0	2.0	1.5	2.0	2.0	2.0	Medium
Latvia	2.0	1.5	1.5	2.0	1.0	2.0	High
Lithuania	2.0	2.0	2.0	2.0	2.0	2.0	Medium
Luxembourg	2.0	2.0	2.0	2.0	2.0	2.0	Medium
Malaysia	1.0	1.5	0.5	1.0	2.0	2.0	Medium
Malta	2.0	1.5	1.5	2.0	0.0	2.0	High
Netherlands	1.5	2.0	1.0	1.0	2.0	2.0	Medium
New Zealand	2.0	1.5	2.0	2.0	2.0	2.0	Medium
Norway	2.0	2.0	2.0	2.0	2.0	2.0	Medium
Poland	2.0	2.0	1.5	2.0	2.0	2.0	Medium
Portugal	2.0	1.5	1.5	2.0	1.0	2.0	Medium
Romania	2.0	2.0	1.5	2.0	2.0	2.0	Medium
Russia	1.5	1.5	0.5	2.0	1.0	2.0	Medium
Saudi Arabia	1.5	1.5	0.5	2.0	2.0	2.0	Medium
Singapore	1.5	2.0	1.0	1.0	2.0	2.0	Medium
Slovakia	2.0	1.5	1.5	2.0	0.0	2.0	High
Slovenia	2.0	1.5	1.5	2.0	1.5	2.0	Medium
South Korea	2.0	1.5	1.5	2.0	2.0	2.0	Medium
Spain	2.0	2.0	2.0	2.0	2.0	2.0	Medium
Sweden	2.0	2.0	1.5	2.0	2.0	2.0	Medium
Thailand	1.5	1.0	1.0	1.0	1.0	2.0	Medium
Türkiye	0.5	0.5	0.0	0.0	2.0	1.0	Medium
Ukraine	0.5	0.5	0.0	0.0	1.5	1.0	Low
United Arab Emirates	2.0	2.0	1.5	2.0	2.0	2.0	Medium
United Kingdom	1.5	1.5	1.0	1.0	2.0	2.0	Medium
United States	2.0	2.0	2.0	2.0	2.0	2.0	Medium
Vietnam	1.5	0.5	1.0	0.0	2.0	2.0	Medium

Table 2. First-pass six-axis scores for 53 countries, derived from the recorded fields (0 to 2 per axis; a dash means not assessed). Reviewer reconciliation pending.

5. Reading one row: France

France illustrates how to read a row. Relevance and impact are established: ANSSI publishes a three-phase transition roadmap (Phase 1 hybridisation now, Phase 2 mandatory hybridisation not earlier than 2025, Phase 3 standalone post-quantum not earlier than 2030), and the NIST baseline is adopted with mandatory hybridisation (ANSSI, 2023, verified). Coherence and governance quality are established through the EU instruments and ANSSI's mandated lead under SGDSN. Effectiveness sits lower because the milestones are framed as not-earlier-than windows and a dedicated measurement duty is not recorded. The verification step corrected one inherited figure: the earlier profile cited fixed visa deadlines of 2027 and 2030, which the ANSSI roadmap does not state, and which should be corrected in the companion report and the regulatory layer.

Part C. Build brief for Claude Code

This is a developer-ready specification for the credibility instrument that shows the six axes on each country profile. The design principle is that playfulness comes from interaction and discovery, not effects: a small precision instrument the visitor operates and compares, in the monochrome style, that never adds a second colour to the map. Every score on screen links to its source, and an unassessed axis is shown empty.

C1. Data shape

Add an optional evaluation object to each country profile (data/profiles/<ISO3>.json), mirror it in Notion, and type it in src/content.config.ts as a nullable field, so profiles without it render an empty gauge. Shape: evaluation maps each of the six criteria (relevance, coherence, effectiveness, efficiency, governance, impact) to an object with score (number 0 to 2, or null for not assessed), confidence (High, Medium or Low), and subScores, an array of { key, label, score (0, 1, 2 or null), reason, sourceUrl }. The loader in content.config.ts reads it from a JSON-encoded text property or from the profile file; the zod schema marks it nullable.

C2. Component

Create src/components/CredibilityGauge.tsx as a React island. Render an SVG (viewBox 0 0 240 240, width 100 per cent). Six axes at 60-degree spacing from the centre, fixed order with relevance at the top and the rest clockwise. Each axis is a thin hairline spoke to the maximum radius; the score plots a point at (score / 2) of that radius; the six points join into a polygon stroked thin in ink and filled in the country's coordination-bloc colour at low alpha, with the bloc colour also on the outer ring as the single accent (read the colour from POSTURE_META in src/lib/process.ts). Axis labels in Schibsted Grotesk sit outside each spoke, with the numeric score in Spline Sans Mono. An axis with a null score renders as a dashed grey spoke with the label 'not yet assessed' and no point; an axis whose confidence is Low renders its point and segment at reduced opacity.

C3. Interaction

On mount the gauge assemblies: the spokes draw with stroke-dashoffset, then the points scale in, over about 1.2 seconds, eased with var(--ease). Hovering or focusing a spoke opens a small panel showing that criterion's sub-scores, each with its 0 to 2 value, its one-line reason, a link to the cited source, and the confidence. A compare control, a select listing the other countries, overlays the chosen country's polygon as a ghosted dashed outline in its own bloc colour, with a two-line legend. Each spoke is a focusable control; Enter or Space toggles its panel.

C4. Accessibility and reduced motion

The svg carries role img with a title and a desc that summarise the country's profile. Each spoke is a button with an aria-label of the form 'Relevance: 2.0 of 2, established'. Values are always shown as text, never by colour alone, and the focus ring uses var(--focus-ring). Under prefers-reduced-motion the gauge renders in its final assembled state with no draw-on or flip, and the sub-score panel is shown inline rather than on an animated reveal.

C5. Integration, tokens and acceptance

Mount the gauge in ProfilePanel, inside src/components/AtlasMap.tsx, directly below the Regulatory basis block, under a new section labelled Governance credibility. Reuse the existing tokens (--ink, --paper, --hairline, --ink-muted, --radius) and fonts (--font-instrument, --font-mono, --font-reading), and the bloc colours from process.ts. Add no new dependency; the gauge is pure SVG and React state. Acceptance: the six axes render from the evaluation data with unassessed axes greyed; hover and keyboard both reveal the sub-scores with a working source link and the confidence; the compare control overlays a second country; colour is never the sole carrier of a value; reduced motion is respected; and the site builds with astro build.

C6. Generative prompt (mood and concept)

Minimal editorial diagram, ink on warm off-white paper. A six-spoke radial gauge, like a precision seismograph, six fine axes from a centre each filled to a different level, one restrained colour accent on the outer ring, the rest monochrome. Clean, scientific, quietly playful through precision rather than effects. Thin strokes, generous whitespace, a small mono label on each axis. Flat 2D, high contrast, no gradients, no glow. Square.

References

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